

Securing Industrial OT with Data Diode

Operational Technology (OT) networks control safety-critical physical processes. Increasing connectivity to enterprise IT networks introduces significant cyber risk.

Key Challenges:

- Industrial protocols lack strong native security
- Firewalls rely on software policies and configuration
- Compromise of IT systems can lead to command injection into OT

Strategic Objective:

Design an FPGA-based hardware data diode to enforce strict OT → IT unidirectional communication.

OT Network Analysis

Characteristics of Industrial OT Networks

- Composed of PLCs, SCADA systems, sensors, and actuators
- Designed for deterministic, real-time operation
- Prioritize availability and stability over flexibility
- Use legacy protocols such as Modbus TCP(request-response based) and UDP
- Sensitive to delay, jitter, and packet loss

Security Insight:

OT protocols were originally designed for isolated environments, not modern interconnected networks.

Strong segmentation is mandatory.

OT-Side Research

OT Architecture & Simulation Study

- Studied industrial OT network architecture and communication models
- Analyzed Modbus TCP request-response behavior
- Evaluated OpenPLC for realistic PLC simulation
- Researched SCADA and industrial traffic generation tools
- Identified architectural incompatibilities with strict one-way enforcement

Critical Finding:

Standard polling models must be redesigned for unidirectional architectures.

This defines the OT-side adaptation requirement.

Proposed Secure Architecture

Hardware-Enforced Secure Monitoring Model

OT Devices → FPGA Data Diode → Digital Twin → Monitoring Stack/IT Devices

Architectural Principles:

- FPGA enforces physical one-way data flow
- No RTL or physical reverse path
- Digital twin replicates OT state safely
- IT systems never access real PLC devices

Security Outcome:

Zero risk of command injection from IT into OT.

Hardware becomes the trust boundary.

Implementation Roadmap (OT-Side Execution)

Phase 1 – Virtual OT Deployment

- Deploy OpenPLC runtime environment
- Develop controlled ladder logic
- Enable Modbus TCP communication
- Generate realistic OT traffic patterns

Phase 2 – Traffic & Protocol Analysis

- Capture and analyze packet size, timing, and frequency
- Study polling intervals and response structure
- Evaluate behavior under constrained (one-way) conditions

Objective: Establish baseline OT communication behavior before FPGA integration.

Integration & Validation Strategy

Phase 3 – FPGA Integration Testing

- Connect OT simulation to FPGA diode
- Verify strict OT → IT data transfer
- Confirm zero reverse signal path
- Inject malformed and stress traffic

Phase 4 – Performance & Security Validation

- Measure latency introduced by diode
- Evaluate packet drop
- Test continuous load stability
- Validate digital twin state consistency

Final Goal:

Demonstrate a secure, stable, hardware-enforced OT monitoring architecture suitable for industrial deployment.

Future Scope – Advanced OT Use-Case Simulations

Expanding Beyond Basic Sensor Simulation

- Extend simulation from simple sensor data to full industrial machine environments
- Model complete OT systems including motors, actuators, and control logic
- Simulate real-time industrial workflows under unidirectional constraints

Example Use Cases

- **CNC Router Machine as a representative OT system**
 - Real-time motor control signals
 - Position feedback loops
 - Telemetry extraction without command injection
- Industrial conveyor systems
- Robotic arms and automated assembly lines, Motor-driven manufacturing equipment

Long-Term Goal

- Validate FPGA data diode across diverse industrial scenarios
- Enable secure telemetry extraction for monitoring and analytics
- Support predictive maintenance and digital twin environments
- Demonstrate scalability for real industrial deployment